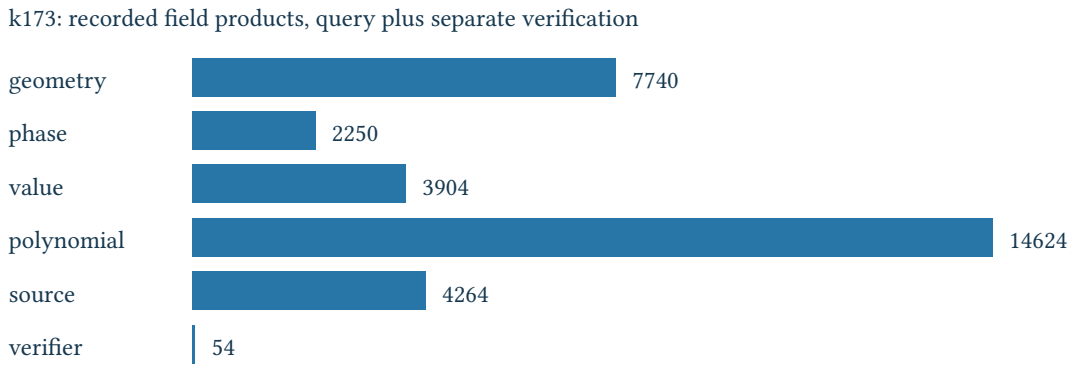
Native recovery arithmetic accounting

Prime-field follow-up · 2026-11-03

Seven reversibly instrumented modules now report setup, reusable preparation, query and independent verification separately. All eight recovery receipts match the unchanged native reference. Three checks pass once; 56 earlier tests are filtered.



Findings

For k173, polynomial work records 14,624 products and 320 inversion calls. Every polynomial inversion operand is one. Geometry records 7,740 products; phase 2,250; value division 3,904; source recurrence and normalization 4,264. Verification separately records 54 products and 11 inversions. Query allocator requests total 599,560 bytes in this build.

What the counters mean

Unit inversions are a subset of inversion calls, not extra work to add. Scalar operations modulo p minus one are separate from source field operations and decoder arithmetic modulo r. Nested stage, thread isolation and allocator forwarding checks pass. Requested allocation bytes include realloc requests; they are not live or peak memory. Optimizer/toolchain behavior can change observed allocations. No timing weights or speed comparison are assigned.

Next construction

Add a monic/linear remainder path and reduce quotient temporaries while retaining the product/remainder tree. Preserve the full original cohort for later validation. Integer work, retained memory, coefficient synthesis and isolated timing remain separate accounting obligations.

Provenance

Exact transformation map, frozen hashes, compiler, commands and logs accompany receipt.json. Two packaging-only generator failures are preserved; the sole native build/test attempt passed. See DERIVATION.md and the Rust GlobalAlloc contract for definitions and limits.